SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	K NAVYA	**Reg. No.:**	192371022
Date:	15/10/2026	**Duration:**	60 minutes

Code of Conduct

I K NAVYA(192371022) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>*.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

AGILE SPRINT PLANNING
AND USER STORY WORKSHOP

Smart Scholarship and Financial Aid Management Platform

Prepared using Agile / Scrum practices

1. Project Vision and Industry Context

Educational institutions administer scholarships and financial assistance schemes that differ in eligibility rules, required evidence, deadlines, review stages and payment procedures. When information is scattered or rules are difficult to interpret, students can miss funding opportunities and administrators spend time checking applications manually.

Vision: Build a secure, transparent and student-friendly platform that publishes scholarship opportunities, identifies potentially eligible students, supports application and document submission, and enables authorized staff to verify, approve and track awards and disbursements.

Project Objectives

- Maintain a central catalogue of scholarships, eligibility criteria, required documents and deadlines.
- Match student profiles against configured criteria and explain that a match is preliminary, not a guarantee of award.
- Support application submission, document verification, approval workflow and status tracking.
- Notify students about deadlines and important status changes.
- Record disbursements and generate accurate operational reports.

Scope and Success Measures

Initial release scope: role-based access, scholarship creation and publication, student profile, basic eligibility matching, application submission, document checklist/status, and an auditable application status. Later increments add richer approval rules, payment integration, analytics and advanced notifications.

Measure	Proposed target / evidence
Opportunity visibility	Published scholarships show eligibility summary, required documents and deadline.
Application completeness	Required fields and documents are checked before submission.
Transparency	Student can view current status and missing items.
Administrative efficiency	Staff can find and review submitted applications in one queue.
Data quality	Every status change and disbursement record is associated with an application and actor.

Assumptions

One institution is the first deployment; institutional staff configure rules; eligibility matching is advisory; financial disbursement is recorded by authorized users rather than initiated through a bank integration in Sprint 1. Privacy, least-privilege access, secure document storage and auditability are cross-cutting requirements.

2. Stakeholders, Roles and Epics

Stakeholder / role	Interest and responsibility
Students / applicants	Discover opportunities, understand requirements, apply, upload evidence and track status.
Scholarship administrator	Create and publish schemes, configure criteria/deadlines, manage the catalogue.
Document verifier	Check evidence, mark verified/rejected, and request corrections with a reason.
Selection / approval committee	Review eligible applications and record approve/reject decisions.
Finance officer	Record award amount, payment date, reference and disbursement status.
Institution management	Monitor demand, approvals, pending cases, awards and disbursement reports.
IT administrator / security team	Manage roles, access, backups, retention and security controls.
Product owner and Scrum team	Set value priorities, refine requirements, build, test and demonstrate increments.

Epics and Example User Stories

Epic	User story (As a… I want… so that…)	Priority
E1: Scholarship catalogue	As an administrator, I want to create and publish a scholarship with a deadline and requirements, so students can find accurate opportunities.	P1
E2: Student profile	As a student, I want to maintain my academic and basic eligibility details, so the platform can assess potential matches.	P1
E3: Eligibility matching	As a student, I want to see scholarships that appear to match my profile, so I can focus on relevant opportunities.	P1
E4: Applications	As a student, I want to complete and submit an application, so my request can enter the review process.	P1
E5: Document verification	As a verifier, I want to mark documents as verified or needing correction, so evidence can be checked consistently.	P2
E6: Approval workflow	As a committee member, I want to record a decision and reason, so outcomes are traceable.	P2
E7: Notifications	As a student, I want deadline and status reminders, so I can act before time runs out.	P2
E8: Disbursement and reports	As a finance officer, I want to record payments and view summaries, so awards can be reconciled and reported.	P3

Personas and key needs

A first-year student needs plain-language eligibility guidance; an administrator needs configurable rules without coding; a verifier needs a clear document queue; and a finance officer needs dependable award and payment records. The design should use accessible language, visible status labels and clear reasons when an application cannot proceed.

3. Product Backlog and Acceptance Criteria

Prioritization uses MoSCoW-style value ranking (P1 = must-have for the first usable increment; P2 = important next; P3 = later). Dependencies are considered before pulling work into a sprint.

ID / story	Priority	Dependency	Acceptance criteria (Given–When–Then)
US-01 Create/publish scholarship	P1	Role access	Given an authorized administrator enters valid title, description, deadline, criteria and required documents, when they publish, then the scholarship appears in the catalogue with those details. Given required fields are missing or the deadline is invalid, when they save/publish, then validation identifies the errors.
US-02 Student profile	P1	Authentication / data model	Given a student is signed in, when they save required profile fields, then the profile is stored and can be edited. Given a required field is absent or invalid, then the system explains what must be corrected.
US-03 Eligibility match	P1	US-01, US-02	Given published scholarships and a completed profile, when matching runs, then the system lists scholarships whose configured rules pass and shows the key matched criteria. If a required criterion fails or data is missing, the scholarship is not labelled eligible; results are described as preliminary.
US-04 Submit application	P1	US-01, US-02; document checklist	Given an open scholarship and a complete application, when the student submits, then a unique application reference and Submitted status are created. If required fields/documents are missing or the deadline has passed, submission is blocked with a clear explanation.
US-05 Verify documents	P2	US-04, staff role	Given a verifier opens a submitted application, when they mark a document Verified or Needs correction and add a reason where needed, then the status, actor and timestamp are saved and visible to authorized users.
US-06 Approve/reject	P2	US-05	Given required verification is complete, when an authorized reviewer records Approve or Reject with any required reason, then the application status and decision audit record are updated. Unauthorized users cannot decide.
US-07 Deadline/status alerts	P2	Scholarship deadlines and contact preference	Given a student has a relevant application or potential match, when a configured reminder/status event occurs, then a notification is queued/sent and logged; duplicate reminders for the same event are suppressed.
US-08 Record disbursement/report	P3	US-06, finance role	Given an application is approved, when finance records amount, date and reference, then a disbursement entry is stored. Reports summarize applications, decisions and payments for the selected period.

Backlog ordering rationale

US-01 and US-02 establish the core data needed for matching. US-03 and US-04 deliver direct student value and validate the main workflow. Document verification and approval depend on submitted applications, while disbursement depends on an approval decision. Notifications and reports are valuable but follow the basic workflow so the team does not automate an unstable process.

4. Sprint 1 Plan and Story Point Estimates

Sprint length: 2 weeks (10 working days). Team assumption: 4–6 people across development, testing and design, with a Product Owner available for clarification. Story points use relative Fibonacci estimates (1, 2, 3, 5, 8, 13), discussed through Planning Poker. Points represent complexity, uncertainty and effort—not hours.

Sprint Goal

Deliver a usable first slice in which an administrator can publish a scholarship and a student can maintain a profile, view preliminary matching opportunities and submit a complete application with a traceable reference.

Selected Sprint Backlog

Story	Points	Why this estimate?	Sprint 1 outcome
US-01 Publish scholarship	5	Moderate form validation, criteria/document fields, persistence and role checks.	Admin can create, validate and publish a scholarship.
US-02 Student profile	3	Small set of profile fields, edit/save flow and validation.	Student can save and update profile data.
US-03 Eligibility match	8	Rule evaluation, missing-data handling, matching display and boundary testing.	Student sees preliminary matches with understandable criteria.
US-04 Submit application	8	Application form, required-field/document checks, deadline guard, unique reference and status.	Complete application can be submitted and tracked.

Planned total: 24 story points. This is a planning assumption, not a guarantee; actual capacity should be calibrated using the team's past velocity. If the team is new, treat this as a forecast and protect the Sprint Goal by reducing scope rather than lowering quality.

Actionable Development Tasks

Story	Tasks (indicative)
US-01	Confirm fields and rules with Product Owner; design scholarship schema; build create/edit form; validate dates and required fields; implement publish/list view; enforce administrator permission; add unit and acceptance tests.
US-02	Define student profile fields; implement profile persistence/API; build profile form; validate inputs; apply access controls; test save/edit and invalid data.
US-03	Translate criteria into supported rule types; implement matching service; handle missing/incomplete profile data; show match explanation and preliminary disclaimer; test pass/fail/boundary cases.
US-04	Define application schema and status transitions; build application form; check required fields/documents and deadline; generate unique reference; persist Submitted status; test duplicate submit, closed deadline and incomplete application.

Dependencies and risk controls

Agree on supported eligibility rule types before coding. Define the meaning of each status early. Use synthetic student data in development, restrict access to uploaded documents, validate file type/size, and avoid exposing sensitive data in logs. External email/SMS and bank integrations are not Sprint 1 dependencies.

5. Sprint Execution, Definition of Done and Deliverables

Suggested Sprint Cadence

When	Activity	Output
Day 1	Sprint Planning: confirm goal, capacity, dependencies and acceptance criteria.	Committed Sprint Backlog and task owners.
Daily (15 min)	Daily Scrum: inspect progress toward goal, surface blockers and adapt plan.	Updated board and blocker actions.
Days 2–8	Design, build, peer review and test in small increments; refine upcoming stories.	Potentially shippable working increment.
Day 9	End-to-end testing, accessibility/security checks, bug fixing and demo preparation.	Release candidate meeting Definition of Done.
Day 10	Sprint Review with stakeholders, then Retrospective.	Feedback and prioritized improvements for next sprint.

Definition of Done (DoD)

- Code is peer-reviewed, merged and follows agreed coding standards.
- Acceptance criteria pass; unit and integration tests pass; key workflow has end-to-end coverage.
- Input validation, role authorization and safe error handling are implemented.
- No known critical/high-severity defects remain; unresolved issues are documented.
- Sensitive student information is not exposed in logs; access is limited to authorized roles.
- User-facing labels and validation messages are understandable; basic accessibility checks are complete.
- Feature is deployed to the agreed test environment and demonstrated to the Product Owner.

Expected Sprint Deliverables

Deliverable	Completion evidence
Scholarship catalogue foundation	Authorized administrator can create and publish a scholarship; invalid data is rejected.
Student profile	Student can save/edit profile and receive validation feedback.
Preliminary eligibility matching	Matching results are based on configured rules and explain key criteria; no award is promised.
Application submission	A complete application submitted before deadline receives a unique reference and Submitted status.
Quality and project artefacts	Test results, updated backlog/board, known-issues list and sprint review feedback.

Risks, Review and Next Sprint

Key risks include ambiguous eligibility rules, inaccurate student data, privacy exposure, late requirement changes and overcommitting story points. Mitigations are rule-owner sign-off, clear missing-data behaviour, least-privilege access, early demonstrations and capacity-based planning. Sprint Review questions: Can staff publish a valid scheme? Can students understand a match? Can an incomplete or late application be prevented? Does each submitted application have a traceable reference?

Likely Sprint 2 candidates: document upload and verification queue (US-05), approval/rejection workflow (US-06), and deadline/status notifications (US-07), reordered after stakeholder feedback and dependency review.

Conclusion

This plan delivers the core scholarship-to-application journey first, prioritizing student access and administrative control. Agile review and feedback will validate the assumptions before the team expands into verification, approvals, disbursement and reporting.

AGILE SPRINT PLANNING AND USER STORY WORKSHOP

Intelligent Procurement Request and Approval System

Prepared using Agile / Scrum practices

1. Project Vision and Industry Context

Organizations purchase equipment, software, office supplies, services, and other resources through procurement processes. Traditional email-based and paper-based approval workflows can lead to delays, limited visibility, difficulty comparing supplier quotations, and challenges in tracking purchase requests.

Vision: Build an intelligent, secure, and transparent procurement management platform that enables employees to submit purchase requests, managers to review requests, procurement teams to collect and compare supplier quotations, and authorized personnel to approve purchases and generate purchase orders.

Project Objectives

- Enable employees to create and submit digital purchase requests.
- Establish configurable approval workflows based on purchase requirements.
- Manage supplier information and quotation details.
- Compare supplier quotations based on price, delivery time, and other criteria.
- Track purchase approvals and rejection reasons.
- Generate purchase orders for approved requests.
- Monitor procurement status throughout the purchasing lifecycle.
- Maintain complete procurement histories and audit records.
- Provide role-based access for employees, managers, procurement staff, and administrators.
- Generate procurement analytics and operational reports.

Scope and Success Measures

Initial release scope: user authentication, role-based access, digital purchase request creation, manager review, basic approval workflow, request status tracking, and procurement history. Later increments will introduce supplier quotation management, quotation comparison, purchase order generation, automated notifications, and advanced analytics.

Measure	Proposed target / evidence
Request submission	Employees can submit valid purchase requests digitally.
Approval transparency	Each request displays its current status and assigned approval stage.
Quotation management	Procurement staff can record and retrieve supplier quotations.
Supplier comparison	Quotations can be compared using price, delivery time, and other configured criteria.
Purchase tracking	Approved requests can be linked to purchase orders and tracked.
Auditability	Each major status change and approval decision records the responsible user and timestamp.
Reporting	Authorized users can view procurement summaries and pending-request reports.

Assumptions

The first release will be deployed within a single organization. Employees, managers, procurement officers, finance approvers, and system administrators will have separate permissions. Approval rules and spending limits

will be configured by authorized administrators. Supplier quotations and purchase orders will initially be managed within the platform, without requiring external ERP or accounting integrations. Security, data validation, access control, and auditability are cross-cutting requirements.

2. Stakeholders, Roles and Epics

Stakeholders and Responsibilities

Stakeholder / Role	Interest and Responsibility
Employees / Requesters	Create purchase requests, provide specifications, attach supporting documents, and track request status.
Department Managers	Review requests, verify business requirements, and approve or reject requests within their authority.
Procurement Officers	Manage supplier details, collect quotations, compare offers, and prepare purchase orders.
Finance Approvers	Review budget availability, costs, and financial compliance before authorizing purchases where required.
Procurement Manager	Oversee procurement operations, resolve exceptions, and monitor purchasing performance.
Suppliers	Provide quotations, pricing details, delivery schedules, and product or service information.
System Administrator	Manage users, roles, permissions, approval rules, and system configuration.
Organization Management	Monitor procurement spending, approval delays, supplier performance, and operational reports.
Product Owner and Scrum Team	Prioritize requirements, develop features, test functionality, and demonstrate sprint increments.

Epics and Example User Stories

Epic	User Story	Priority
E1: User and Role Management	As an administrator, I want to assign roles to users so that only authorized users can perform specific actions.	P1
E2: Purchase Requests	As an employee, I want to submit a digital purchase request so that my purchasing needs can be reviewed efficiently.	P1
E3: Approval Workflow	As a manager, I want to review and approve or reject requests so that purchases follow organizational policies.	P1
E4: Supplier Management	As a procurement officer, I want to maintain supplier details so that supplier information is available during purchasing.	P2
E5: Quotation Management	As a procurement officer, I want to record supplier quotations so that available offers can be evaluated.	P2
E6: Quotation Comparison	As a procurement officer, I want to compare supplier offers so that the organization can make informed purchasing decisions.	P2
E7: Purchase Order Generation	As a procurement officer, I want to generate purchase orders for authorized purchases so that suppliers receive clear order details.	P2
E8: Notifications and Tracking	As an employee, I want to receive status updates so that I know the progress of my purchase request.	P2
E9: Procurement Analytics	As a management user, I want to view procurement reports so that I can monitor expenditure, delays, and supplier performance.	P3

Personas and Key Needs

An employee needs a simple request form and clear status information. A department manager needs a review queue with request details and approval history. A procurement officer needs an organized quotation comparison interface and purchase order generation. A finance approver needs visibility into costs and budget compliance. Management needs reliable procurement reports and spending summaries. The system should provide understandable status labels, clear rejection reasons, traceable decisions, and easy access to relevant procurement records.

3. Product Backlog and Acceptance Criteria

Prioritization uses a MoSCoW-style approach: P1 represents must-have functionality for the first usable increment, P2 represents important subsequent functionality, and P3 represents later enhancements. Dependencies are considered before assigning work to a sprint.

ID / Story	Priority	Dependency	Acceptance Criteria (Given–When–Then)
US-01: User and Role Management	P1	Authentication	Given an administrator is authorized, when they assign a role to a user, then the permissions associated with that role are applied. Unauthorized users cannot access restricted functions.
US-02: Create Purchase Request	P1	US-01; request data model	Given an employee is signed in, when they enter valid item details, quantity, estimated cost, department, and justification, then the system saves the request and generates a unique reference. Missing or invalid mandatory fields produce validation messages.
US-03: Submit Purchase Request	P1	US-02	Given a valid draft request exists, when the employee submits it, then its status changes to Submitted and it becomes available to the appropriate reviewer. Duplicate submission is prevented.
US-04: Review and Approve Request	P1	US-03; approval permissions	Given a request is awaiting review, when an authorized manager approves or rejects it, then the decision, reviewer, timestamp, and applicable reason are recorded. Unauthorized users cannot make the decision.
US-05: Track Request Status	P1	US-03, US-04	Given an employee has submitted a request, when they open the request details, then they can view its current status and relevant status history. They cannot view another employee's restricted information without permission.
US-06: Manage Suppliers	P2	US-01	Given a procurement officer enters valid supplier information, when they save it, then the supplier record is available for quotation management. Invalid or

			incomplete mandatory details are rejected.
US-07: Record Supplier Quotations	P2	US-06; approved procurement request	Given a procurement officer has obtained supplier offers, when they record the supplier, quoted price, delivery time, and validity period, then the quotation is linked to the relevant request and stored for review.
US-08: Compare Quotations	P2	US-07	Given two or more quotations are recorded, when the procurement officer opens the comparison view, then the system displays comparable prices, delivery times, and other configured criteria. The officer can identify a preferred offer and record the selection rationale.
US-09: Generate Purchase Order	P2	Approved request; quotation selection	Given a purchase is fully authorized and the supplier selection is recorded, when the procurement officer generates a purchase order, then the system creates a unique order number and stores the supplier, items, quantities, prices, and order date. Unauthorized or unapproved purchases cannot generate an order.
US-10: Notifications	P2	Request and approval events	Given a request is submitted or its status changes, when a configured notification event occurs, then the relevant user is notified through the configured channel and the event is logged.
US-11: Procurement Reports	P3	Request, approval, and order records	Given an authorized management user selects a reporting period, when they generate a report, then the system summarizes requests, approval outcomes, pending cases, purchase orders, and available expenditure data for that period.

Backlog Ordering Rationale

US-01 establishes secure access and authorization. US-02 and US-03 create the basic request submission process, while US-04 and US-05 establish approvals and visibility. Supplier management and quotation handling depend on the request workflow. Quotation comparison depends on stored supplier offers, and purchase order generation depends on the required approvals and supplier selection. Notifications and analytics are scheduled after the core workflow has been established.

4. Sprint 1 Plan and Story Point Estimates

Sprint length: 2 weeks (10 working days). Team assumption: 4–6 people across development, testing, and design, with a Product Owner available for clarification. Story points use relative Fibonacci estimates (1, 2, 3, 5, 8, 13), discussed through Planning Poker. Points represent complexity, uncertainty, and effort rather than hours.

Sprint Goal: Deliver a usable first version in which authorized users can access the system according to their roles, employees can create and submit purchase requests, managers can record approval decisions, and employees can track request status.

Selected Sprint Backlog

Story	Points	Why This Estimate?	Sprint 1 Outcome
US-01: User and Role Management	5	Requires authentication integration, role assignment, permission checks, and access testing.	Users access only the functions permitted by their roles.
US-02: Create Purchase Request	5	Requires form design, data validation, persistence, and request reference generation.	Employees can create and save valid purchase requests.
US-03: Submit Purchase Request	3	Requires submission validation, status transitions, and reviewer assignment.	Valid requests enter the review workflow.
US-04: Review and Approve Request	8	Requires decision handling, authorization checks, approval records, and status history.	Authorized managers can approve or reject submitted requests.
US-05: Track Request Status	3	Requires status display, history retrieval, and access restrictions.	Employees can view their request progress.

Planned total: 24 story points. This is a planning assumption, not a guarantee. Actual sprint capacity should be calibrated using the team's previous velocity. If the team is new, treat this as a forecast and protect the Sprint Goal by reducing scope rather than lowering quality.

Actionable Development Tasks

Story	Indicative Tasks
US-01	Define user roles and permissions; design the user data model; implement authentication and role checks; build user administration functions; test authorized and unauthorized access.
US-02	Define request fields; design the database schema; build the purchase request form; implement validation and persistence; generate unique request references; test invalid inputs.
US-03	Define request status transitions; implement submission logic; prevent duplicate submissions; assign the appropriate reviewer; test valid and invalid submission scenarios.
US-04	Design the approval workflow; implement manager review screens; add approve/reject actions; store reviewer, timestamp, and decision reason; test authorization and decision history.
US-05	Design status labels and history views; retrieve request status and audit records; apply access controls; test status visibility and restricted access.

Dependencies and Risk Controls

Define request statuses and approval permissions before implementing the workflow. Establish mandatory request fields and validation rules with the Product Owner. Use test data during development and ensure that employees cannot modify approval decisions or access restricted requests. The initial sprint should not depend on external supplier systems, ERP integrations, or financial integrations.

5. Sprint Execution, Definition of Done and Deliverables

Suggested Sprint Cadence

When	Activity	Output
Day 1	Sprint Planning: confirm the Sprint Goal,	Agreed Sprint Backlog and task owners.

	capacity, dependencies, and acceptance criteria.	
Daily (15 min)	Daily Scrum: inspect progress, identify blockers, and adapt the plan.	Updated task board and blocker actions.
Days 2–8	Design, development, code review, testing, and backlog refinement.	Working increments of procurement functionality.
Day 9	End-to-end testing, security checks, defect fixing, and demonstration preparation.	Release candidate meeting the Definition of Done.
Day 10	Sprint Review with stakeholders, followed by the Sprint Retrospective.	Stakeholder feedback and prioritized improvements.

Definition of Done (DoD)

- Code is peer-reviewed, merged, and follows agreed coding standards.
- User stories satisfy their acceptance criteria.
- Unit, integration, and relevant end-to-end tests pass.
- Input validation and role-based authorization are implemented.
- Purchase requests and approval decisions are stored correctly.
- Status changes and approval actions include the required audit details.
- No known critical or high-severity defects remain; unresolved issues are documented.
- Sensitive procurement information is accessible only to authorized users.
- User-facing labels and validation messages are understandable.
- The increment is deployed to the agreed test environment and demonstrated to the Product Owner.

Expected Sprint Deliverables

Deliverable	Completion Evidence
User and role management	Authorized users can access permitted functions; restricted access is tested.
Digital purchase request form	Employees can create and save requests; invalid submissions are rejected.
Request submission workflow	Valid requests receive unique references and Submitted status.
Approval workflow	Authorized managers can record approval or rejection decisions with traceable records.
Procurement status tracking	Employees can view the status and relevant history of their requests.
Quality and project artifacts	Test results, updated backlog, sprint board, known-issues list, and review feedback are available.

Risks, Review, and Next Sprint

Key risks include unclear approval rules, incomplete purchase specifications, unauthorized access, inconsistent quotation data, and overcommitting sprint capacity. Mitigation measures include obtaining approval-rule sign-off, validating mandatory fields, enforcing least-privilege access, recording audit events, and demonstrating working functionality early.

Sprint Review questions:

- Can employees create and submit valid purchase requests?
- Can only authorized managers approve or reject requests?
- Can employees track their request status?
- Are approval decisions and status changes recorded accurately?
- Can the team demonstrate the end-to-end request and approval workflow?

Likely Sprint 2 candidates: Supplier management (US-06), supplier quotation recording (US-07), quotation comparison (US-08), and notification support (US-10), subject to stakeholder feedback and dependency review.

Purchase order generation can follow once quotation selection and the complete approval requirements are implemented.

Conclusion

The Intelligent Procurement Request and Approval System aims to improve purchasing efficiency by replacing fragmented manual processes with a centralized digital workflow. The proposed Agile approach prioritizes secure request submission, approval tracking, and status visibility before expanding into supplier quotation management, quotation comparison, purchase order generation, and procurement analytics. By maintaining complete procurement histories, enforcing role-based permissions, and providing transparent approval records, the system can reduce paperwork, improve supplier evaluation, support better purchasing decisions, and reduce procurement delays. Iterative development and stakeholder feedback will help ensure that the final system meets the organization's procurement requirements.